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Deep Learning–Based Decomposition Methods for Financial Volatility Forecasting

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stocks for high-frequency volatility forecasting. The empirical evaluations were performed using metrics including RMSE, MAE, and RMSPE. The paper evaluates three mode decomposition methods: Empirical Mode Decomposition (EMD), Variational Mode Decomposition (VMD), and complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) for volatility forecasting. The empirical results evidence the superior performance of the VMD utilizing the LSTM over other decomposition techniques utilizing LSTM and LSTM variants. This research underscores the effectiveness of hybridizing LSTM with advanced mode decomposition techniques, highlighting their potential to enhance volatility forecasting in financial markets.

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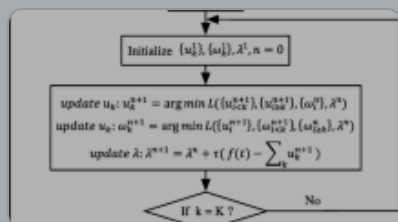
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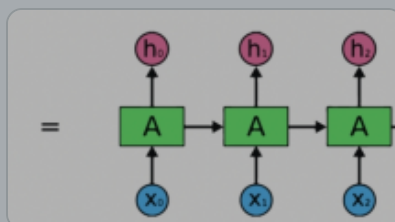
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